a FlexLifeFi K–12 program

GRADE 7 · LESSON PLANS

Ten Lessons, Ready to Teach

Standards, materials, procedure, differentiation — pick up Monday.

Year theme: Know Your Customer

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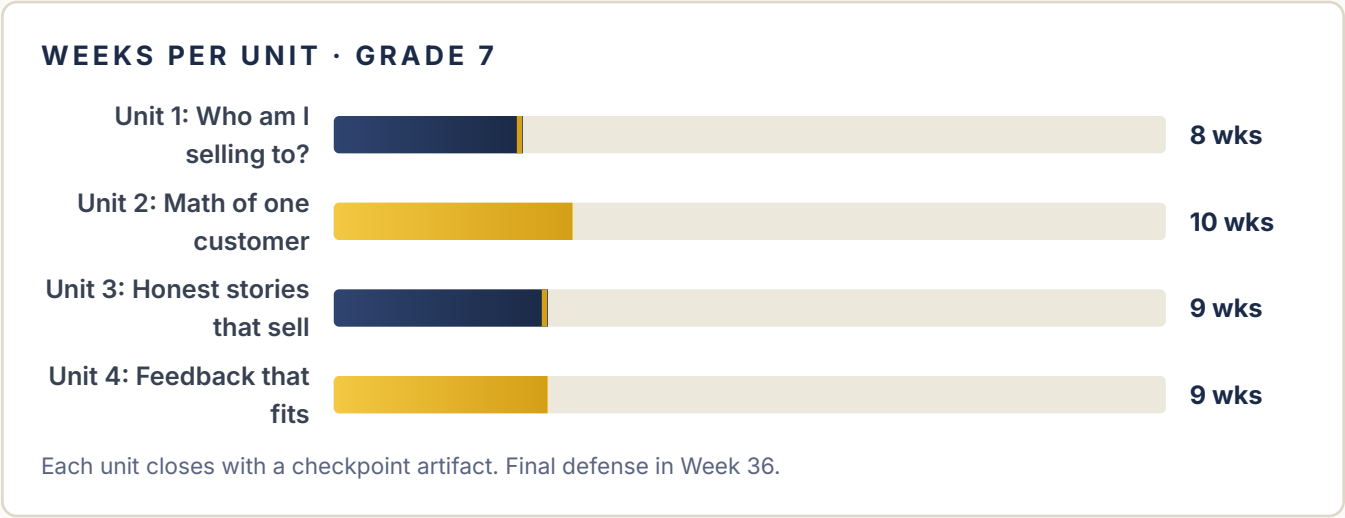
YEAR-AT-A-GLANCE

Four Units. Thirty-Six Weeks. One Capstone.

The year is built as four units, each ending with a public-facing artifact that feeds the rolling Capstone. The bars show how the weeks are spent — roughly equal, with a slight lean toward the math-heavy unit because per-customer math is the load-bearing skill.



36 WEEKS

4 UNITS

10 LESSONS

8 CASH VIDEOS

GRADE 7 · LESSON PLANS

Ten Lessons, Ready to Teach

Customers are evidence, not assumptions.

Ten fully fleshed lessons spanning the year. Each lesson is built to pass the picks-up-Monday test: a teacher with no prior prep should be able to read this Sunday night and teach it Monday morning. Standards are cited at the benchmark level. Differentiation notes are specific and actionable. Vocabulary is included for each lesson. Time blocks default to 50 minutes (single middle-school period) unless noted; longer blocks are flagged.

Lessons are sampled across the year so the full arc is visible:

1. (Week 1) Cash V1 launch + Day 1 Customer Assumption
2. (Week 4) Open vs. Leading Questions — Interview Practice
3. (Week 6) Coding Patterns from Interview Transcripts
4. (Week 8) Persona Pack — Checkpoint 1 Studio Day
5. (Week 11) Variable Cost per Customer — From Receipt to Per-Unit
6. (Week 13) Contribution Margin — What's Left to Pay the Rent?
7. (Week 21) Persuasion vs. Manipulation — Drawing Your Line
8. (Week 25) Running a 3-Day Mini A/B Test
9. (Week 32) Reading the Survey — Signal vs. Noise
10. (Week 36) Final Capstone Defense — Defense Day Itself

Lesson 1 — Day 1 Customer Assumption

"Don't sell. Show."

— CASH

Grade band: 6-8 **Time:** 50 minutes **Unit:** 1 — Who am I selling to? **Topic area:** Customer awareness · entrepreneurship launch

Objectives

By the end of this lesson, students will be able to:

- Write a "Day 1 customer assumption" — a single sentence naming who their hypothetical Capstone customer is and what they want.
- Distinguish between assumption and evidence using two concrete examples.

Standards alignment

- **Jump$tart Standard 6 (Financial Decision Making), Benchmark 8-2:** Identify ways that consumers may avoid making poor decisions. (Applied to entrepreneurs avoiding poor customer guesses.)
- **CCSS-ELA W.7.4:** Produce clear and coherent writing in which the development, organization, and style are appropriate to task.
- **CASEL Self-Awareness:** Identifying one's own assumptions.

Materials

- Cash Video #1 ("Customers are evidence, not assumptions") — 2:30. (If no projector, use the printed transcript.)
- One Capstone Journal (notebook) per student.
- Two index cards per student.
- A 2-foot strip of butcher paper labeled "Assumption" on one end and "Evidence" on the other, taped to the floor or wall.
- A list of 10 sample statements, half assumptions, half evidence (printed on cards or projected). Examples: "My friends will buy it" (assumption); "Three of my friends said they'd pay $5 for it on Tuesday" (evidence); "Older kids love this kind of thing" (assumption); "Two 8th graders chose ours over the other in a side-by-side test" (evidence).

Vocabulary

- **Assumption** — something you think is true without checking.
- **Evidence** — something you observed, heard, or measured that supports or contradicts what you think.
- **Customer** — a specific person who would receive the product or service in exchange for something (money, attention, time).

Procedure

1. Engage — Cash Video + One Question (~7 min)

Play Cash Video #1. As students watch, post the prompt on the board:

> *"What's a time you assumed something about another person and turned out to be wrong?"*

Pair-share for 2 minutes. Cold-call two students to share. (You're surfacing the human muscle that the rest of the year will train onto customers.)

2. Explore — Assumption-Evidence Continuum (~12 min)

Distribute one card to each student. (If you have 25 students, multiple students may get the same card.) Read the rules:

> *"I'm going to read 10 statements. After each one, walk to the spot on this line that matches: pure assumption all the way left, solid evidence all the way right, somewhere in the middle if it depends."*

Read each statement aloud, pause, allow movement. After 3–4 statements, pick two students at different points and ask "what made you pick that spot?" Don't correct — let students hear each other's reasoning. Continue through all 10.

After the last one, ask the closing question: "What's the difference between an assumption and evidence in one sentence?" Two volunteers; you sharpen the wording on the board.

3. Explain — Vocabulary + the Year's Norm (~8 min)

Define the three vocabulary words. Add to Vocabulary Spine.

State the year's first norm: "In this room, when you say something about a customer, you'll be asked: which people, and how do you know? You'll ask each other this. I'll ask. You'll get good at answering."

Tell them they will write a Day 1 Customer Assumption and revisit it in Week 8 to see how wrong they were. Frame as a feature, not a bug — being wrong on Day 1 is normal and useful.

4. Apply — Write the Day 1 Assumption (~15 min)

In the Capstone Journal, students write three things:

1. The kind of business they're imagining (one sentence).
2. **The Day 1 Customer Assumption.** Format: "I think my customer is [who] who wants [what] because [why]."
3. One thing they're least sure about in their assumption.

Confer with 5 students while others write — read their sentence, ask one sharpening question.

5. Close — Pin and Predict (~8 min)

Each student writes the Day 1 Customer Assumption on an index card with their name. Collect them. Tell them you will hand them back in Week 8 and they'll see how the customer changed once evidence got involved. Project this final exit prompt:

> *"Predict: in Week 8, how wrong will your Day 1 assumption be? On a scale of 1 (totally right) to 5 (completely wrong)? Write your number on the back of the card."*

Collect cards. End class.

Differentiation

- **ELL:** Provide the sentence frame on a sticky note: "I think my customer is **_ who wants because __**." Allow native-language brainstorming with a peer; require English on the index card. Pre-teach assumption and evidence with cognates if applicable (Spanish: suposición, evidencia).
- **IEP/504:** Allow a verbal version recorded on phone (with student transcribing later). Provide a printed list of the 10 statements during the continuum activity so students can follow without having to track audio. Extended writing time on the Day 1 Assumption — okay to finish in the next session.
- **Advanced learners:** Push for specificity in the assumption — "older kids" is too broad. Ask "which older kids? what age, where do they hang out, what's the moment they'd buy?" Have them write a second sentence: what would have to be true for the customer to buy. This sets up Unit 2's per-customer math.

Assessment

Formative: Watch the continuum activity for which students place statements correctly. Confer notes during the writing time. **Summative:** Day 1 Customer Assumption is filed; will be compared to Week 8 Persona for one of the Capstone's reflection prompts.

Closure

Same as #5 above.

Homework / Extension

Read the page in the Student Activity Book titled Customer-First Stories — three short founder origin stories — and circle a sentence in any one of them where the founder talks about who their customer was at the start.

Teacher notes

- The continuum activity is the engine of this lesson. If your room can't accommodate movement, project the line on the board and use sticky notes. Don't skip it.
- Some students will struggle to name a Capstone topic on Day 1. That's fine — name a "kind of business" they're curious about. They'll lock the topic in Week 2.

- A student says "I don't have a business idea." Coach: "What's something you're good at, or curious about? What's something annoying that nobody fixes?" Read the Skills + Hobbies + Needs framing if helpful.

Lesson 2 — Open vs. Leading Questions

Grade band: 6-8 **Time:** 50 minutes **Unit:** 1 — Who am I selling to? **Topic area:** Customer interviewing · evidence-gathering

Objectives

By the end of this lesson, students will be able to:

- Identify whether an interview question is open, closed, or leading, with a reason.
- Rewrite three leading questions into open questions.

Standards alignment

- **CCSS-ELA SL.7.1:** Engage effectively in collaborative discussions, building on others' ideas and expressing their own.
- **CCSS-ELA W.7.7:** Conduct short research projects to answer a question.
- **CASEL Social Awareness:** Perspective-taking — letting the customer's view come through.

Materials

- 12 printed cards, each with one interview question (mix of open, closed, leading).
- Capstone Journal.
- Sample interview transcript (1 page, provided in program assets) — used as the "before" example.
- Wall chart with three columns: OPEN, CLOSED, LEADING.

Vocabulary

- **Open question** — a question that lets the answerer say what they actually think; usually starts with what, how, why, tell me about.
- **Closed question** — a yes/no or short-answer question. Useful for confirming a fact, not for understanding.
- **Leading question** — a question that signals the answer the asker wants. "You'd buy this, wouldn't you?"

Procedure

1. Engage — Bad Interview (~7 min)

Read aloud the sample bad interview transcript: an interviewer leads the interviewee through 5 leading questions and "confirms" their idea is great. Stop after each question and ask the class to thumbs-up if it's a good question, thumbs-down if not.

Reveal: every question was leading. The interview is useless evidence.

2. Explore — Sort the Cards (~10 min)

Pass out the 12 cards. In pairs, students sort them into Open / Closed / Leading. Walk pairs to the wall chart to post their sort. After 5 minutes, read the wall chart aloud — where did the class agree? Where did they disagree? Discuss the disagreements.

Sample cards (mix into your set):

- "Tell me about the last time you bought a snack at school." (open)
- "Do you buy snacks at school?" (closed)
- "You think the snacks here are too expensive, right?" (leading)
- "What was annoying about your last lunch line wait?" (open)
- "How long was your wait yesterday?" (closed — but useful)
- "Wouldn't you love a faster line?" (leading)
- "Walk me through a typical Tuesday at lunch." (open)
- "Which would you prefer, A or B?" (closed but neutral)
- "Most people think A is better, what do you think?" (leading)

3. Explain — The Three Types (~5 min)

Define the three. Surface a clean rule: Open questions get you stories; closed questions get you facts; leading questions get you nothing useful.

4. Apply — Rewrite the Leading Questions (~18 min)

Each student picks 3 leading questions from a list of 6 (provided in the activity book) and rewrites each as an open question that gets at the same information. Examples:

- "You'd buy this, right?" → "What would make you decide to buy something like this?"
- "Don't you hate when they raise the prices?" → "When did you last notice a price change? How did you react?"

After 10 minutes of writing, students swap with a partner and check each other's rewrites with two questions: - Does the new question still get at what we wanted to learn? - Could the answerer disagree freely without disappointing the asker?

Revise based on partner feedback.

5. Close — Build Tomorrow's Interview (~10 min)

Each student writes the first 3 questions of their own Capstone interview script. They must include at least 1 open and 1 closed; no leading allowed. Confer briefly with as many students as possible. Collect at end.

Differentiation

- **ELL:** Provide a question stem bank ("Tell me about... What was it like when... How did you decide... What surprised you about..."). Allow drafting in native language first; finalize in English. Pair with a strong English speaker for the partner check.
- **IEP/504:** Reduce the rewrite count from 3 to 2. Provide pre-printed sentence stems. Allow voice memo of the 3 interview questions instead of writing.
- **Advanced learners:** Add a "follow-up question" requirement — for each open question, write a possible follow-up that probes a story detail. Introduce the term probe informally.

Assessment

Formative: Watch the wall-chart sort. Read the rewrites during conferring.
Summative: The 3-question script feeds into the interview activity in Week 5.

Closure

Same as #5.

Homework / Extension

Listen to one short podcast interview (any) and identify three open questions. Bring the timecodes (or the questions transcribed) to Wednesday.

Teacher notes

- Watch for the trap of "open" questions that are still leading: "Tell me about how much you love our school's food." Coach: open in form, leading in tone. The test is can the person answer freely?
- A student will say "I'm scared to interview a real person." Validate, then practice with a peer first. Most kids fear it less after one try.

Lesson 3 — Coding Patterns from Interview Transcripts

Grade band: 6-8 **Time:** 60 minutes (block — or 2 × 30-minute periods) **Unit:** 1 — Who am I selling to? **Topic area:** Customer evidence · qualitative analysis (kid version)

Objectives

By the end of this lesson, students will be able to:

- Highlight three different "patterns" in a transcript using a 3-color coding system.
- Tally the patterns and write 1–2 sentences describing what the customers most consistently said.

Standards alignment

- **CCSS-ELA RI.7.1:** Cite several pieces of textual evidence to support analysis.
- **CCSS-Math 7.SP.A.1:** Understand that statistics can be used to gain information about a population by examining a sample.
- **CASEL Self-Awareness:** Recognizing the gap between assumptions and evidence.

Materials

- Each student's three interview transcripts (homework from Week 5).
- Three highlighter colors per student (yellow = problem; blue = workaround; pink = surprise).
- Tally sheet (provided in activity book).

Vocabulary

- **Pattern** — something that shows up more than once across different sources.
- **Code** — a short label you put on a piece of evidence so you can count it.
- **Sample** — the small group you actually talked to; not necessarily the whole population.

Procedure

1. Engage — Three Yellow Highlights (~8 min)

Project a 1-page sample transcript (provided). Demo: read it once, then highlight in yellow every place where the interviewee mentions a frustration or problem. Count them out loud. Notice that some are repeated, some are unique. Pose: "If you were going to build something to help this person, which yellow highlight would you start with?"

Pair-share. One pair shares.

2. Explore — Three-Color Code (~20 min)

Students apply the three-color code to one of their own transcripts: - **Yellow** — problem or frustration the customer mentions. - **Blue** — workaround the customer is currently using. - **Pink** — anything that surprised you.

Then they tally counts on the Tally Sheet for that one transcript. (Tally sheets have rows for each color and a column for "what was said.")

After 12–15 minutes, repeat with a second transcript, then a third. Most students will get through 2 in this block; finish the 3rd in independent work time.

3. Explain — Pattern vs. One-Off (~8 min)

Mini-lesson: a pattern shows up across at least 2 of 3 transcripts. A one-off shows up in just one. Both are interesting. Patterns are more reliable. Make the rule: "Treat one-offs as hypotheses to check. Treat patterns as evidence."

4. Apply — Write the Pattern Sentence (~15 min)

In the Capstone Journal, students write:

1. **My yellow pattern:** "Across [N] interviews, [N] people mentioned [problem]." (1–2 sentences.)
2. **My blue pattern:** "When facing this, customers currently [workaround]."
3. **My biggest surprise:** "I didn't expect [thing]. This makes me think [implication]."

Confer with as many students as possible.

5. Close — Wall of Patterns (~9 min)

On a sticky note, each student writes their yellow pattern and posts it to the class wall. Class quickly walks the wall (1 minute). Two volunteers share what surprised them about the wall. End.

Differentiation

- **ELL:** Allow native-language interviews if the customer was an ELL family member; permit transcripts in original language but require the pattern sentence in English. Provide sentence frames for the pattern sentence. Pre-highlight a "model" transcript as a worked example.
- **IEP/504:** Reduce to one transcript coded in class; finish remaining at home (or in resource room). Provide pre-printed Tally Sheet with the three colors labeled. Larger highlighters if fine motor is a concern.
- **Advanced learners:** Push toward second-order coding — within yellow, are there sub-themes (price, time, quality)? Have them propose a 4th color that captures something interesting they're noticing.

Assessment

Formative: Walk during coding; check that students aren't highlighting wholesale (every yellow defeats the purpose). Read pattern sentences during conferring.
Summative: The pattern sentences feed Persona v2 in Week 7.

Closure

Same as #5.

Homework / Extension

Bring all three transcripts, fully coded, to Friday's Showcase.

Teacher notes

- A student returns transcripts that are too short or unrevealing. Coach: "What was their longest answer about? Code that one fully." Even a short transcript yields one pattern.
- Some students realize their interview script was bad (leading questions). Validate the realization — that's evidence too. Allow a 4th do-over interview if time permits.

Lesson 4 — Persona Pack Studio Day (Checkpoint 1)

"Most adults skip this. You won't."

— CASH

Grade band: 6-8 **Time:** 80 minutes (block) — Checkpoint day **Unit:** 1 — Who am I selling to? **Topic area:** Customer persona · synthesis

Objectives

By the end of this lesson, students will be able to:

- Produce a Persona Pack consisting of: a 1-page persona, a problem statement, and a "what I changed since Day 1" reflection.
- Defend the Persona Pack in a 3-minute peer share, citing at least 2 pieces of evidence from interviews.

Standards alignment

- **Jump$tart Standard 6 (Financial Decision Making), Benchmark 8-3:** Apply systematic decision making to identify the role of emotion in making financial decisions. (Applied to customer empathy.)
- **CCSS-ELA W.7.4:** Produce clear and coherent writing.
- **CCSS-ELA SL.7.4:** Present claims and findings with pertinent descriptions.
- **NGSS-ETS MS-ETS1-1:** Define the criteria and constraints of a design problem with sufficient precision.

Materials

- Persona one-page template (provided).
- Day 1 Assumption index cards (returned from Lesson 1).
- Capstone Journal with all coded transcripts.
- Wall space for the class persona gallery.
- Rubric (Customer Research rubric from the Rubric Bank).

Vocabulary

(All previously introduced — recycle, no new terms today.) Reinforce: persona, segment, problem statement, evidence.

Procedure

1. Engage — Hand Back Day 1 Cards (~5 min)

Hand back the Day 1 Customer Assumption cards. Quietly. Ask students to read their own and write one sentence in the Journal: "What's the biggest difference between this and what I now think?"

2. Build the Persona (~30 min)

Students work independently with the Persona one-page template. Sections:

- **Persona name** (a real first name, not real person — e.g., "Marcus" or "Priya").
- **One-line summary** (e.g., "12-year-old who plays after-school soccer and is always hungry by 4:00").
- **Demographic** (age range, grade, neighborhood pattern).
- **Daily routine** (what's their Tuesday look like?).
- **Top 3 frustrations** (each backed by a quote from a transcript).
- **Where to reach them** (3 channels — physical or digital).

Confer with each student for 2–3 minutes.

3. Write the Problem Statement (~15 min)

Mini-lesson on the problem statement format:

> *"[Persona name] struggles with [problem] when [context], because [cause]."*

Example, modeled aloud: "Marcus struggles with finding affordable fresh food after soccer practice when most snack options are vending machines, because the school store closes at 3:30."

Each student drafts their own problem statement, revises with a partner, and writes the final.

4. Reflection — What Changed Since Day 1 (~10 min)

Students write the third piece of the Persona Pack:

> *"On Day 1, I assumed my customer was* ***[_]. After interviews, I now think my customer is [].*** *The biggest evidence-based change was* ***[_], because []****."*

Two paragraphs, 4–6 sentences each.

5. Gallery Walk + Peer Share (~15 min)

Tape Persona Packs to wall. Students walk in silence for 5 minutes, leaving sticky-note feedback ("I see..." "I wonder..."). Then 3 students share their Persona Pack in 3 minutes each — 1 minute persona, 1 minute problem statement, 1 minute reflection. Use the rubric criteria as listening structure.

6. Submit (~5 min)

Persona Packs are submitted (digital or paper) for grading using the Customer Research rubric.

Differentiation

- **ELL:** Provide the persona template with each section pre-labeled and a vocabulary glossary on the back. Sentence frames for the problem statement. Allow oral defense in pair-share rather than whole-class share.
- **IEP/504:** Pre-fill student name and date; chunk the worksheet to one section per page. Extended time — finish next class period. Allow audio submission of the reflection.
- **Advanced learners:** Add a "secondary persona" section — who is not this product for, and why? This previews Grade 8's segmentation work. For the reflection, push for "the assumption I had that was most wrong" with a specific causal hypothesis.

Assessment

Summative: The Persona Pack is graded on the Customer Research rubric (Rubric Bank). Anchor papers from prior years (or instructor-built exemplars) are used to calibrate scoring.

Closure

Same as #6.

Homework / Extension

Capstone Journal entry: "What's one thing I still don't know about my customer that I need to figure out by the end of Unit 2?"

Teacher notes

- A student's Day 1 assumption matches their Week 8 persona. This is rare and usually means weak interviewing. Probe: "What did you actually learn? What was a surprise?" If they had no surprise, the interviews were probably leading.
- A student has trouble naming a persona without using a real classmate's name. Frame as fictional but plausible — the persona is a composite, not a person.
- The peer share is high-leverage; protect the time. Even if the gallery walk runs short, do all 3 share-outs.

Lesson 5 — Variable Cost per Customer

Grade band: 6-8 **Time:** 50 minutes **Unit:** 2 — Math of one customer **Topic area:** Per-unit costs · cost reasoning

Objectives

By the end of this lesson, students will be able to:

- Calculate variable cost per unit for a hypothetical product using a sample receipt.
- Distinguish variable cost from fixed cost using 5 example items.

Standards alignment

- **Jump$tart Standard 6 (Financial Decision Making), Benchmark 8-2:** Identify ways consumers may avoid making poor decisions. (Applied to entrepreneurs.)
- **CCSS-Math 7.RP.A.3:** Use proportional relationships to solve multistep ratio and percent problems.
- **CCSS-Math 7.RP.A.1:** Compute unit rates.
- **CEE Standard 1 (Earning Income), Grade 8 Benchmark:** Income from self-employment is not the same as profit.

Materials

- Sample receipt: a fictional bulk supply purchase. E.g., a paint set: "1 case (24 brushes), $36; 1 case acrylic paints (20 tubes), $40; 50 mini canvases, $30."
- A "kid art party" hypothetical: each customer gets 1 brush, 2 paint tubes, 1 canvas.
- Worksheet: 3 sample products, students compute variable cost per customer.
- Calculator (or mental math).

Vocabulary

- **Variable cost** — a cost that goes up when you sell more (more brushes, more paint).
- **Fixed cost** — a cost that stays the same regardless of how many customers you have (a table rental, a website fee).
- **Cost per unit** — what it costs you to make/serve one customer.
- **Unit economics** — the math of one customer, used to figure out the whole business.

Procedure

1. Engage — The Receipt (~5 min)

Project the fictional receipt. Pose: "You're throwing a kid art party. Each kid gets 1 brush, 2 paints, 1 canvas. What does each kid 'cost' you?" Pair-share predictions; collect 3.

2. I do — Modeling the Math (~10 min)

Walk through the math out loud:

- Brush: $36 ÷ 24 = $1.50 per brush; 1 per kid → $1.50.
- Paint: $40 ÷ 20 = $2.00 per tube; 2 per kid → $4.00.
- Canvas: $30 ÷ 50 = $0.60 per canvas; 1 per kid → $0.60.
- **Total variable cost per kid:** $1.50 + $4.00 + $0.60 = **$6.10.**

Surface the rule: variable cost per unit = sum of (item cost ÷ items in package × items used).

3. We do — Group Practice (~10 min)

Together, work a second example: "A friendship-bracelet station: 50 cords for $10, 200 beads for $4. Each kid uses 1 cord and 8 beads."

- Cord: $10/50 = $0.20; 1 each = $0.20.
- Beads: $4/200 = $0.02; 8 each = $0.16.
- **Total VC: $0.36 per bracelet.**

4. You do — Worksheet (~15 min)

Students complete the worksheet with 3 sample products on their own:

- A snack-bag service.
- A custom keychain.
- A 30-minute backyard tutoring session (where the variable cost is photocopies + maybe a small reward sticker).

Then they identify, from a list of 8 items, which are variable costs and which are fixed costs:

- Bag of beads (variable)
- Monthly Etsy fee (fixed)
- Cardstock for thank-you notes per customer (variable)
- Logo design (fixed)
- Box per shipment (variable)
- Photo background prop (fixed)
- Per-customer ad cost (variable; "if I pay $0.50 per click and 1 in 10 clicks buys, my ad cost per customer is $5")
- Annual website domain (fixed)

5. Apply to Capstone (~10 min)

Students compute the variable cost per unit for their own Capstone, in the Capstone Journal. Confer with as many as possible. Collect notebooks at end.

Differentiation

- **ELL:** Provide a reference card with the variable cost formula in plain language. Cognates in Spanish: costo variable, costo fijo. Pair with a strong-math peer for the worksheet section.
- **IEP/504:** Calculator allowed throughout. Worksheet is pre-formatted with rows and columns labeled. Reduce variable/fixed sort from 8 items to 5.
- **Advanced learners:** Add a "bulk discount" twist — what if the brushes are $36 for 24, but $54 for 50? At what number of customers does the bulk pack make sense? Introduces breakeven reasoning ahead of Week 15.

Assessment

Formative: Worksheet performance during the You do; conferring on Capstone application. **Summative:** The per-customer cost feeds the Pricing Pack at Checkpoint 2 (Week 18).

Closure

Exit ticket: "In your own words, what's the difference between variable cost and fixed cost?" One sentence.

Homework / Extension

Find one real product you could "make" or "serve" and write its variable cost per unit. Bring the math Wednesday.

Teacher notes

- Watch for ""home"-supplies" supplies. A student says "but my mom buys the supplies." The math of the business doesn't care who pays — the cost is real. Coach: "If you ran this at scale, you'd buy them. What would they cost?"
- A student says variable cost is the same as price. Distinguish: cost is what you pay to make it; price is what you charge. The gap is contribution margin (next lesson's topic — preview).

Lesson 6 — Contribution Margin: What's Left to Pay the Rent?

Grade band: 6-8 **Time:** 50 minutes **Unit:** 2 — Math of one customer **Topic area:** Contribution margin · pricing math

Objectives

By the end of this lesson, students will be able to:

- Calculate contribution margin per unit (price – variable cost) for a sample product.
- Explain in 2–3 sentences why contribution margin matters more than total revenue for understanding the business.

Standards alignment

- **Jump$tart Standard 6 (Financial Decision Making), Benchmark 8-2:** Identify ways consumers may avoid making poor decisions.
- **CCSS-Math 7.RP.A.3:** Use proportional relationships.
- **CCSS-Math 7.EE.B.4:** Use variables to represent quantities and construct simple equations.
- **CEE Standard 1 (Earning Income), Grade 8 Benchmark:** Profit equals revenue minus expenses.

Materials

- "What's Left to Pay the Rent?" worksheet (provided).
- Last lesson's variable-cost numbers (students should have them in the Capstone Journal).
- Two example businesses (lemonade stand and tutoring) with full numbers.

Vocabulary

- **Contribution margin (per unit)** — what's left from the price after you cover the variable cost. The amount each customer "contributes" toward your fixed costs and profit.
- **Margin** — same idea, often expressed as a percentage of price.

Procedure

1. Engage — Two Lemonade Stands (~7 min)

Project a side-by-side:

	Stand A	Stand B
Price per cup	$2.00	$1.00
Variable cost per cup	$1.80	$0.30
Cups sold	50	30
Revenue	$100	$30

Ask: "Which stand is in a better spot to grow? Why?"

Pair-share. Most students will pick A (more revenue!). Reveal:

- Stand A: contribution per cup = $2 − $1.80 = $0.20. Total contribution = $10.
- Stand B: contribution per cup = $1 − $0.30 = $0.70. Total contribution = $21.

Stand B has more money left over to cover fixed costs and reinvest. Revenue isn't the whole story.

2. Explain — The Definition (~5 min)

Define: **Contribution margin per unit = price per unit – variable cost per unit.** Each customer "contributes" that amount toward your fixed costs and profit. Surface the year's metaphor: "It's what's left to pay the rent."

3. We do — Tutoring Example (~10 min)

Together: a 30-min tutoring session priced at $15. Variable costs: $1 worth of photocopies, a $0.50 sticker reward.

- VC = $1.50 per session.
- CM per session = $15 − $1.50 = $13.50.
- If the tutor does 4 sessions/week, weekly contribution = $54.
- If the tutor's "fixed cost" is a $25/month tutoring app subscription = $6.25/week. Profit = $54 − $6.25 = $47.75/week.

Run a "what if" — what happens if price drops to $10? CM drops to $8.50. Weekly contribution drops to $34. Profit drops to $27.75. Coach: small price changes have outsized profit effects.

4. You do — Apply to Capstone (~20 min)

Students fill in the "What's Left to Pay the Rent?" worksheet for their own Capstone:

- Price per unit (best guess for now).
- Variable cost per unit (from Lesson 5).
- Contribution margin per unit.
- Estimated units per month (best guess).
- Total monthly contribution.
- One example fixed cost.
- Profit estimate.

Confer with as many students as possible. Common stuck point: "what's a fixed cost in my business?" Coach: a website domain ($15/year), a monthly app fee, time-as-cost (introduced at survey level), supplies bought in bulk that don't tie to a single customer.

5. Close — One Sentence Why It Matters (~8 min)

Exit ticket: "Contribution margin matters more than revenue because [____]."

Two volunteers share. End.

Differentiation

- **ELL:** Provide a reference card with the contribution margin formula and a worked example. Pair on the worksheet for the first scenario, independent for the Capstone application.
- **IEP/504:** Calculator allowed. Worksheet has units and labels pre-printed. Permit working through one Capstone scenario fully rather than multiple "what ifs."
- **Advanced learners:** Add a "what if my variable cost drops 10%" scenario and a "what if my price rises 10%" scenario. Compute both. Surface the asymmetry — pricing power affects margin more than cost-cutting at most price points (within reason).

Assessment

Formative: Worksheet performance and Capstone confer. **Summative:** Feeds the Pricing Pack at Checkpoint 2 (Week 18).

Closure

Same as #5.

Homework / Extension

Capstone Journal: write your CM per unit and answer: "If I doubled my price, what's the most that could go wrong?"

Teacher notes

- The lemonade hook is load-bearing. Don't skip it — it dislodges the "more revenue = better" frame.
- A student computes a negative contribution margin (variable cost > price). This is great teaching — they're losing money on every sale. Coach: "Should you sell this? Or change something? Which?"

Lesson 7 — Persuasion vs. Manipulation: Drawing Your Line

"Listening is not a moment. It's a system."

— CASH

Grade band: 6-8 **Time:** 50 minutes **Unit:** 3 — Honest stories that sell **Topic area:** Ethics in marketing · ad analysis

Objectives

By the end of this lesson, students will be able to:

- Identify three persuasion techniques in real ads (scarcity, social proof, authority, urgency, fear).
- Distinguish persuasion from manipulation using a 4-point Fair-to-Customer Check, with a written justification for one borderline case.

Standards alignment

- **CEE Standard 2 (Buying Goods and Services), Grade 8:** People can use information from sellers and outside reviewers to compare products. (Flipped: students learn to be honest sellers.)
- **CCSS-ELA RI.7.8:** Trace and evaluate the argument and specific claims in a text.
- **CCSS-ELA W.7.1:** Write arguments to support claims.
- **CASEL Responsible Decision-Making:** Ethical responsibility.

Materials

- 6 real or realistic ads (printed or projected): a mix — a clearly honest one, a clearly misleading one, and 4 borderline. (Provided in program assets.)
- 4-point Fair-to-Customer Check (provided): clear claim · honest price · real reviews · easy returns.

- Worksheet: ad analysis with a structured response.

Vocabulary

- **Persuasion** — convincing someone using accurate, relevant information.
- **Manipulation** — convincing someone using inaccurate or unfair information, or by exploiting a feeling that doesn't match the facts.
- **Scarcity** — "only 3 left!" Sometimes true, sometimes invented to pressure.
- **Social proof** — "10,000 customers love it!" Sometimes true, sometimes inflated.
- **Authority** — "doctors recommend!" Sometimes real, sometimes paid.
- **Urgency** — "today only!" Sometimes a real deadline, sometimes a fake one.

Procedure

1. Engage — Spot the Trick (~10 min)

Project the 6 ads, one at a time. For each, ask: "What's this ad trying to make you feel?" Pair-share for 30 seconds, share-out 2 students per ad.

After all 6, surface the techniques on the board: scarcity, social proof, authority, urgency, fear. Each technique has a true version and a fake version. Distinguish out loud: "Only 3 left" is true if there are 3 left in stock. It's manipulative if the seller has 300 in a warehouse.

2. Explain — The Fair-to-Customer Check (~7 min)

Introduce the 4 points:

1. **Clear claim** — Is what the product does specific and verifiable?
2. **Honest price** — Is the price you'll pay clearly stated, including fees?
3. **Real reviews** — Are reviews actually from customers (not paid)?
4. **Easy returns** — Can you give it back if it doesn't work for you?

A failure on any of these doesn't necessarily mean the ad is manipulative — but a failure on multiple is a red flag.

3. Apply — Pair Analysis (~15 min)

In pairs, students take the worksheet. Each pair gets 2 of the 6 ads (one clear, one borderline). They run each through the Fair-to-Customer Check and write a 2-sentence verdict: persuasion, manipulation, or borderline-and-here's-why.

4. Discuss — Where's Your Line? (~10 min)

Whole class. Project a borderline case (suggested: an ad for a teen "study supplement" with vague claims and a celebrity endorsement). Run a four-corners activity:

- "Persuasion (it's fine)"
- "Manipulation (it's not fine)"
- "Borderline — leans persuasion"
- "Borderline — leans manipulation"

Students go to the corner. Each corner sends one spokesperson with a 30-second case. Then offer the chance to switch corners. Don't tell students what's right; let the discussion surface the criteria.

5. Close — Your Capstone Pledge (~8 min)

Students write a 3-line pledge in the Capstone Journal:

- In my Capstone, I will not [**_] because [**_].
- In my Capstone, I will [**_] because [**_].
- If a friend told me to "stretch the truth a little" to sell more, I would [____].

End class.

Differentiation

- **ELL:** Provide a printed glossary for the persuasion techniques with cognates (Spanish: escasez, autoridad, urgencia). Allow native-language verdict, with English summary required. Sentence frames: "This ad is __ because __."
- **IEP/504:** Reduce ads from 6 to 4. Simplified Fair-to-Customer Check with checkboxes. Permit verbal pledge instead of written.
- **Advanced learners:** Add a "design challenge" — take the most manipulative of the 6 ads and rewrite the headline + 2 lines of body copy to be persuasive but honest. Defend the rewrite against the original.

Assessment

Formative: Pair analysis worksheets; participation in four corners. **Summative:** Pledge feeds the Messaging Pack at Checkpoint 3 (Week 28).

Closure

Same as #5.

Homework / Extension

Find one ad in the wild this week (TV, online, sign in a window). Run it through the Fair-to-Customer Check. Bring it Friday.

Teacher notes

- Students will want a clean rule. Resist that. Persuasion vs. manipulation lives on a spectrum, and the lesson is the judgment, not the verdict.
- A student names a brand they love as manipulative. Validate the observation; coach toward "what specifically is it doing? Is that fair to customers?"
- Sensitivity flag: ads targeted at teens often involve appearance and body. Have a fall-back set if a flagged ad surfaces accidentally.

Lesson 8 — Running a 3-Day Mini A/B Test

Grade band: 6-8 **Time:** 50 minutes (Day 1 of a 3-day arc; Days 2–3 are 15-minute check-ins) **Unit:** 3 — Honest stories that sell **Topic area:** Marketing experimentation · evidence-based decision

Objectives

By the end of this lesson, students will be able to:

- Design a 3-day A/B test of two marketing messages with a defined success metric.
- Identify one source of bias their test cannot eliminate.

Standards alignment

- **NGSS-ETS MS-ETS1-3:** Analyze data from tests to determine similarities and differences among several design solutions.
- **CCSS-Math 8.SP.A.1:** Patterns of association in bivariate data. (Lite version at 7th grade.)
- **CCSS-ELA W.7.7:** Conduct short research projects.

Materials

- Mini A/B test template (provided).
- Two posters per student/team (or two digital images, depending on test medium).
- Tally sheet: "Message A picks" vs. "Message B picks."
- Whiteboard and markers.

Vocabulary

- **A/B test** — comparing two versions to see which performs better, with everything else held the same.
- **Metric** — the number that tells you who won. (e.g., picks, sign-ups, votes.)
- **Bias** — a way the test could be unfair (the bigger poster, the prettier color, the kid presenting it).

Procedure

1. Engage — Two Snack Posters (~8 min)

Project two example posters for a hypothetical "study snack box":

- **Message A:** "The brain food smart kids actually choose."
- **Message B:** "30 grams of protein. No crash at hour two."

Ask: "Which would make you more likely to buy? Why?" Pair-share. Surface that A leans on social proof + flattery; B leans on specifics + benefit.

Tell them: "If we wanted to know which actually works, what would we have to do?" Surface "test both."

2. Explain — The Setup (~10 min)

Define A/B test. Walk through three rules:

1. **Two messages, one variable changing.** If both posters have different colors AND different words, you don't know which mattered. Hold all else equal.
2. **Define the metric before you start.** "Whichever gets more votes" is fine; whichever "feels better" isn't.
3. **Count.** Tally honestly. Don't fudge.

Walk through the test format: each team will tape both posters on the school wall (or in the classroom, or in a known foot-traffic area), with a small "vote" jar/clipboard under each. Track for 3 days. Whichever has more votes at end of Day 3 is the winner.

Add the bias warning: "You can't eliminate all bias. But you can name yours. If your A poster is on the right and people read left to right, that's a bias. Note it."

3. Design — Build Your Two Posters (~25 min)

Students pair up (or work alone) to build the two posters for their Capstone. The two messages must be:

- About the same product.
- The same length (within 10%).

- Visually similar (same size, same colors, same layout — only the words change).

Confer during build.

4. Close — Predict the Winner (~7 min)

In the Capstone Journal, each student writes:

- The two messages.
- Which they think will win, and why.
- One bias their test can't eliminate.

Tomorrow they post the test and start counting.

Differentiation

- **ELL:** Provide a poster template with placement guides; messages can be drafted in native language, finalized in English. Pair on the test design.
- **IEP/504:** One poster pair per team if doing solo would be overwhelming. Provide pre-printed templates with two text boxes.
- **Advanced learners:** Push toward two messages testing two specific psychological hooks (e.g., social proof vs. concrete-benefit). Predict not just the winner but by how much.

Assessment

Formative: Confer during poster build. **Summative:** The 3-day test feeds the Messaging Pack at Checkpoint 3.

Closure

Same as #4.

Homework / Extension

Day 2 (15-minute check-in): tally votes. Day 3: tally + analysis.

Teacher notes

- Some students will have unequal traffic in the test environment (one wall vs. another). Coach: that's a bias — name it in the analysis. Imperfect tests are fine; honest about imperfection is required.
- If the test ends in a tie or near-tie, coach: that's a real result. Many real A/B tests are inconclusive. The next move is to test a different variable, not to declare a winner.

Lesson 9 — Reading the Survey: Signal vs. Noise

Grade band: 6-8 **Time:** 50 minutes **Unit:** 4 — Feedback that fits **Topic area:** Feedback analysis · qualitative + quantitative reading

Objectives

By the end of this lesson, students will be able to:

- Sort survey responses into signal (worth acting on) and noise (one-off or unclear) using two named criteria.
- Choose one finding from their feedback survey and propose a concrete change to the Capstone in response.

Standards alignment

- **CCSS-Math 7.SP.A.2:** Use data from a random sample to draw inferences about a population.
- **CCSS-ELA RI.7.1:** Cite several pieces of textual evidence.
- **NGSS-ETS MS-ETS1-3:** Analyze data from tests.
- **CASEL Self-Management:** Self-monitoring and adjustment.

Materials

- Each student's collected feedback survey results (from Week 32).
- A "Sample Feedback Survey" example (provided): 12 hypothetical respondents to a snack-box service.
- Signal/Noise sort grid.
- Capstone Journal.

Vocabulary

- **Signal** — feedback that matters and could change something.
- **Noise** — feedback that's a one-off, unclear, or about something you can't or shouldn't change.
- **Theme** — a pattern across multiple responses.

Procedure

1. Engage — One Feedback Story (~7 min)

Tell a 2-minute story (or play a Cash short): a kid runs a custom-keychain service, gets 20 surveys back, and the loudest one says "I HATE the colors." Ask: "What would you do? Change the colors? Ignore the kid? Something else?"

Pair-share. Surface that one loud opinion is rarely a reason to change. Many similar opinions often is.

2. Explain — Two Criteria (~8 min)

Two questions to apply to every piece of feedback:

1. **Frequency.** Did multiple people say something similar? (Theme.)
2. **Actionability.** Could you actually do something about it? (Or is it a wish, like "I wish it were free"?)

If both are yes → signal. If frequency is no but actionability is yes → keep an eye on it.
If actionability is no → noise (or a longer-term aspiration).

3. Sort — Sample Survey Together (~12 min)

Project the 12 sample responses. As a class, sort them into Signal / Borderline / Noise.
Examples:

- "More variety in the snacks." (8 of 12 said something similar) → Signal.
- "I'd like a free version." (1 person, but actionable in the sense of a free trial) → Borderline.
- "The packaging is too plastic." (4 of 12) → Signal (also a values check — see Ethics strand).
- "Make it taste like ramen." (1 person) → Noise.
- "Deliveries are sometimes late." (3 of 12) → Signal.

4. Apply — Sort Your Own Survey (~15 min)

Students sort their own feedback into Signal / Borderline / Noise on the grid sheet.
Confer with as many as possible. Push back on signal claims based on 1 person ("Is that a theme, or a one-off?").

5. Close — One Change, Defended (~8 min)

In the Capstone Journal, each student writes:

- The strongest signal in my feedback was [____].
- Because of this, I'm changing [____].
- Here's how I'll know if the change worked: [____].

End. They'll execute the change in Week 33–34.

Differentiation

- **ELL:** Sentence frames for the journal entry. Allow sorting in pairs. Pre-translate the sample survey responses if helpful.
- **IEP/504:** Reduce sample survey from 12 to 8 responses. Provide a checklist version of the two criteria (frequency: yes/no; actionability: yes/no).
- **Advanced learners:** Identify a quiet signal — something only 1–2 people said but that, if true, would be very important. Defend whether to investigate it. Introduces the concept that not all signal is loud.

Assessment

Formative: Sort sheets and confer notes. **Summative:** The "one change, defended" feeds the Feedback Pack at Checkpoint 4.

Closure

Same as #5.

Homework / Extension

Make the change. Capture in the Capstone Journal: "What I changed and what happened next." Bring Friday.

Teacher notes

- Some students will have very small samples (3–5 respondents). The math of inference doesn't work well at that size — coach to be honest about it: "with this sample, my best guess is X, and I'd want to check more."
- Some students will get harsh feedback. Validate the effort; coach toward "what's the kernel of truth?" Don't let students dismiss real signal because it stings.

Lesson 10 — Final Capstone Defense: Defense Day

"Revenue is loud. Contribution margin is true."

— CASH

Grade band: 6-8 **Time:** 90 minutes (block — or two periods) **Unit:** Capstone synthesis
Topic area: Reflection · public presentation · year-long synthesis

Objectives

By the end of this lesson, students will be able to:

- Deliver a 5-minute Capstone Defense covering: (a) the customer they ended with, (b) their per-customer math, (c) their messaging decision, (d) the change they made because of feedback.
- Respond to two panel questions citing specific evidence from the year's work.

Standards alignment

- **CCSS-ELA SL.7.4:** Present claims and findings, emphasizing salient points in a focused, coherent manner.

- **CCSS-ELA SL.7.5:** Include multimedia components and visual displays.
- **Jump$tart Standard 6 (Financial Decision Making), Benchmark 8-1:** Apply systematic decision making.
- **CASEL Self-Awareness + Responsible Decision-Making.**

Materials

- Each student's full Capstone Journal.
- Defense slides or poster (built in Week 35).
- Defense rubric (Pitch Presentation rubric from the Rubric Bank).
- Panel of 3–5 evaluators (other teachers, parents, older students, community members).
- Timer.

Vocabulary

(All previously introduced — recycle.)

Procedure

1. Engage — The 60-Second Warm-Up (~10 min)

Pair students up. Each gives a 60-second "elevator" version of their Capstone to their partner. Partner asks one question. Repeat with a new partner. (Two rounds total. This is a confidence warm-up before the panel.)

2. Set the Norms (~5 min)

State the day's norms clearly:

- Each student gets 5 minutes for the defense + 3 minutes for panel Q&A.
- The panel is curious, not adversarial. They are looking for evidence and reflection, not for "is this a great business?"
- The audience listens silently and writes one "I noticed..." and one "I wondered..." per presenter.

Show the rubric (Pitch Presentation) one more time so students know what's being scored.

3. Defenses (~60 min, ~12–15 students at 8 min each; longer block or split day for more)

Each student delivers the 5-minute defense with the structure:

- **The customer (1 min):** "I learned my customer is X, and here's what's specific about them — quote, quote, quote."
- **The math (1 min):** "My contribution margin is X. Here's why I priced it where I did."

- **The message (1 min):** "I tested two messages. A won/B won/it tied because of X. My final message is Y."
- **The change (1 min):** "Because of feedback, I changed Z. Here's what happened next."
- **What I learned about myself (1 min):** "The biggest thing I got wrong on Day 1 was X. The biggest thing I'd do differently next year is Y."

After each: 3 minutes of panel Q&A. Timer keeper enforced strictly.

4. Audience Reflection (~5 min)

Each audience member completes a 2-question card:

- One thing I noticed across many defenses today: __.
- One thing I'd want to learn more about: __.

Collect.

5. Final Cash Reflection (~10 min)

Play Cash Video #8 — the year-end "Customer-First" summary. Students write one final Capstone Journal entry:

- The biggest thing I learned this year was [____].
- One thing I'd tell next year's Grade 7 student is [____].
- I'm proud of [____].
- I'm going to keep working on [____].

6. Close (~5 min)

Hand back Day 1 Customer Assumption cards. Students re-read theirs. Two volunteers share what 12 weeks (and 12 more) of evidence did to it.

Differentiation

- **ELL:** Allow visuals to carry more weight in the defense — props, posters, charts. Sentence frames for Q&A: "I think [____] because [evidence]." Permit a same-language peer to translate panel questions on the spot if needed.
- **IEP/504:** Allow the defense to be delivered in shorter chunks if 5 min is overwhelming — 1-minute mini-segments with a 30-second pause. Permit a printed script. Audio-record version submitted in advance is also acceptable.
- **Advanced learners:** Add a "what would have to change for me to actually launch this?" segment. Push for a 1-page written addendum identifying the highest-leverage next step.

Assessment

Summative: The defense is graded on the Pitch Presentation rubric. Capstone Journal across the full year is a separate portfolio grade using the Reflection Writing rubric.

Closure

Same as #6.

Homework / Extension

(End of year — none required.)

Teacher notes

- Panel members need a 10-minute briefing before defenses start. Cover: rubric, norms, what to ask for evidence, what not to ask (no judgments on whether the venture would "really work"). Provide them with 5 sample questions: "What was the biggest surprise from your interviews? What would you do if your customer changed her mind tomorrow? Walk me through your contribution margin again. What was the hardest piece of feedback to hear? If you had one more month, what's the first thing you'd change?"
- Some students will be very nervous. The 60-second warm-up matters. So does the audience-card structure — it makes peers active listeners, not silent watchers.
- The Day 1 card return at the end is the year's emotional payoff. Don't skip.

Senior expert upgrade pack — Grade 7 misconceptions and worked examples

Common Misconceptions table

Kid belief	What's actually true	How to coach
"I can imagine my customer."	Imagined customers are guesses. Real customers leave evidence. The fastest way to be wrong about a business is to imagine the customer instead of meeting one.	"Tell me one quote from a real interview. If you don't have one, you don't have a customer yet — you have a hypothesis."
"Demographics ARE the persona."	Demographics are the easiest part. Behaviors and frustrations are the persona. "12-year-old female" is a demographic. "12-year-old who plays soccer Tuesday and Thursday and forgets breakfast on game days" is a persona.	"Add three behaviors. Drop two demographics."
"Higher revenue = better business."	Revenue is loud. Contribution margin is true. A high-revenue business with negative contribution margin is losing money on every customer.	Use the Stand A vs. Stand B example. "Which one made more profit? Why?"

Kid belief	What's actually true	How to coach
"An A/B test result of 8 vs. 6 is a clear winner."	With small samples, an 8-vs-6 split is barely signal. Could be noise. The test was useful for direction, not certainty.	"What would you do differently if you ran it again with twice the sample? What if 8 vs. 6 reverses?"
"Honest claims sell less than clever ones."	Honest claims sell long. Clever claims sell short. The customer comes back to the honest one.	"Name a brand you stopped trusting after a misleading ad. How fast did you stop buying?"
"If a customer says 'I'd buy it,' that's evidence."	Words are not behavior. Buying is behavior. Putting money down is the only evidence that someone is willing to pay.	"What did they actually do? Did they give you a deposit, sign up, share, or just nod?"

Third contribution margin worked example (deepening Lesson 6)

The original lesson uses Stand A vs. Stand B and a tutoring example. Add a third worked example with a counterintuitive result.

Worked example — The high-priced low-margin trap

A custom-keychain service. The student has been told "premium products charge premium prices."

	Plan A: $5 keychain	Plan B: $15 keychain
Variable cost	$1 (basic supplies)	$9 (engraved metal, premium box)
Contribution margin	$4	$6
Estimated sales/month	40	8
Monthly contribution	$160	$48

Coach: "The premium price has a higher per-unit margin (good) but lower volume and higher costs eat into total contribution. The cheap version makes more money. There's no rule. There's only the math."

ROUND 2 PATCH — Common Misconceptions (Grade 7)

Common kid-belief	Reality	Coaching move
"Revenue means were profitable."	Revenue is cash in. Contribution margin is whats left after the variable cost of serving that customer. Revenue can be high and contribution margin negative.	Use the Stand A vs. Stand B counterintuitive reveal. Stand A: $100 revenue, $80 variable cost → $20 contribution. Stand B: $50 revenue, $20 variable cost → $30 contribution. Stand B is more profitable per customer.
"A persona is a fictional character I made up."	A persona is a fictional character built from real evidence. Quote, observation, count — the persona is grounded in interviews, not imagination.	Show two persona cards. One has zero quotes; one has 3 quotes from interviews. Ask which is the real persona.
"If I interviewed my best friend, thats one customer interview."	Your best friend is biased toward agreeing with you. Customer interviews need people who would tell you the truth — which often means strangers, not friends.	Walk through the Mom Test posture. Show 3 sample interview questions and which ones lead the witness vs. uncover truth.
"If 5 people said yes, the customer base is solid."	5 people is a small sample. The pattern matters more than the count. Are they the same kind of yes, or 5 different kinds of yes?	Run a 5-quote analysis. Code each yes for the reason. If the reasons are inconsistent, the customer base isnt solid yet.
"Day 1 customer assumptions are usually right."	Day 1 assumptions are usually 30-50% wrong. Thats why we file them and re-open them at Day 30. The point is to see your own thinking change.	At Day 30, open the Day 1 letter aloud. Count assumptions still true vs. revised. Affirm the revisions.

ROUND 2 PATCH — Multi-Class Branching Scenario (Grade 7)

"Two pricing teams meet, both lemonade, different per-customer math." A 3-day branching scenario.

- **Day 1.** Team A: $1.50/cup, $0.30 variable cost = $1.20 contribution per cup. Team B: $1.00/cup, $0.20 variable cost = $0.80 contribution per cup. Whos more profitable per customer? (A.) Whos more profitable in total? (Depends on volume.)

- **Day 2.** Each team writes a 1-page persona of their target customer. Team A: "premium fair-goer, willing to pay for quality." Team B: "high-volume kid market, price-sensitive."
- **Day 3.** Each team defends their per-customer math against the others persona. End with a debrief: which math is right? (Both, for their customer.)

This branching practices BAVC (compare two businesses), MFL (per-customer revenue + contribution margin), CBM (persona), LTO (team huddle), LRCI (Day 1 vs. Day 30 reasoning).

ROUND 2 PATCH — Teacher Move Set (Grade 7)

- **Student treats a 1-customer interview like a customer base.** Say: "One interview is one persons truth. Whats the pattern across 5? If you only have 1, run 4 more before drawing conclusions."
- **Student writes a persona with no quotes.** Say: "Your persona has no evidence. Lets add 3 quotes from your interviews. Where are they?"
- **Student says "my persona is just like me."** Say: "Are you the customer? Lets check. Would you pay for this product? If yes, build the persona of you. If no, find a different person."
- **Student opens the Day 1 letter and finds it embarrassing.** Say: "Embarrassing means your thinking changed. Thats the win. Lets count: which assumptions held? Which moved?"
- **Student designs a customer feedback survey with 10 questions.** Say: "10 is too many. Pick 3 — one open, two closed. Whats the most useful question?"

ABOUT THE HOST

Cash, in five words

curious · warm · honest · patient · specific

"I'm Cash. I don't promise wealth. I don't promise success. I promise that the moves — interview, math, message, listen — will make you a clearer thinker about money no matter what you end up doing."

— CASH

What Cash does

- **Asks questions that point to evidence.** Cash never tells; Cash asks what your evidence says.
- **Names rules sparingly.** "An assumption is something you think is true without checking. Evidence is something you observed, heard, or counted."
- **Tells short, made-up-but-realistic stories.** 30–60 seconds, shows don't tell.
- **Celebrates the move, not the win.** Revision, honesty, and evidence — not revenue.
- **Hedges deliberately.** probably, often, sometimes — never absolutes about complex topics.

What Cash never does

- Flatter ("amazing!", "great job!"), promise wealth, moralize about money.
- Reference real brands or living celebrities as endorsements.
- Surface family financial details. (Hypothetical scenarios only.)
- Pretend the world is fair. Cash is honest about systemic factors without becoming a politics class.

> *When in doubt, work one of these into the script: Which people, and how do you know? · Being wrong on Day 1 is the deal. · Revenue is loud. Contribution margin is true. · A little less wrong, every week.*

STRAND OUTCOMES

Eight Strands. One Year. Verifiable Outcomes.

Cash to Company runs the same eight strands K-12. Each year deepens the demonstration. The Grade 7 version of each strand is below.

Code	Strand	Grade 7 demonstrated outcome
BAVC	Business Awareness and Value Creation	Identify a business problem, propose a value-creation hypothesis, and revise it with customer evidence.
PSIG	Problem Solving and Idea Generation	Generate, screen, and refine ideas; document why one was chosen and the others were not.
MFL	Money and Financial Literacy	Compute revenue, variable cost, contribution margin, and breakeven for a real product or service.
CBM	Customers, Branding, Marketing	Build a persona from interview evidence; design honest messaging; A/B test two claims.
OSO	Operations, Systems, Organization	Design a simple weekly operating cadence with named roles, handoffs, and a feedback loop.
LTO	Leadership, Teamwork, Ownership	Run and document a team meeting; resolve one disagreement using shared evidence.
ESL	Ethics, Safety, Legal Awareness	Identify the line between persuasion and manipulation; flag one borderline claim and rewrite it honestly.
LRCI	Launch Readiness and Continuous Improvement	Pilot, measure, and document at least three iterations with rationale and what changed.

STANDARDS ALIGNMENT

Six Benchmarks This Year

Standards are cited at the benchmark level. Honest gaps are named when they exist. The Capstone artifact addresses each one.

Benchmark	What students must demonstrate
Jump$tart 6 (Financial Decision Making) 8-2	Identify ways consumers may avoid making poor decisions; apply to entrepreneurs avoiding poor customer guesses.
CCSS-ELA W.7.4	Produce clear and coherent writing in which the development, organization, and style are appropriate to task.
CCSS-ELA SL.7.1	Engage effectively in a range of collaborative discussions.
CASEL Social Awareness	Recognize the perspectives of others; demonstrate empathy.
C3 D2.Eco.1.6-8	Explain how economic decisions affect well-being of individuals, businesses, society.
Jump$tart 1 (Spending & Saving) 8-2	Demonstrate ability to use number skills to manage money.

HOW TO READ THIS TABLE

Each lesson cites at least one of these benchmarks. Each Capstone checkpoint addresses two or more. The Teacher Guide includes a per-lesson standards crosswalk.

"Standards aren't a cage. They're a list of moves you're going to be glad you can make. Pick one. Practice it. Move on."

— CASH

CAPSTONE RUBRIC (4-POINT)

How We Score the Capstone

The Capstone artifact is scored on four traits, each on a 4-point scale. Honest, specific, decision-oriented work earns the high marks. Generic or assumption-laden work earns the low ones.

Trait	1 — Emerging	2 — Developing	3 — Proficient	4 — Advanced
Evidence quality	Mostly assumptions; few or no quotes.	2-3 interview quotes; partial evidence.	5+ direct quotes; varied sources; coded patterns.	Triangulated quotes plus behavioral observation; counter-evidence acknowledged.
Per-customer math	Confused costs; no contribution margin.	CM computed but with errors; one price scenario.	Correct CM and breakeven; clear pricing rationale.	CM under multiple price scenarios; LTV-lite reasoning.
Honest messaging	Vague or misleading claims.	Some specifics; some unsupported claims.	All claims supported; clean ethics check.	A/B-tested copy with rationale; resists clever-but-borderline phrasing.
Feedback system	One-time check; no documentation.	Ad-hoc feedback; partial revisions.	1-week feedback loop with 3 documented changes.	Sustained loop plus signal-vs-noise analysis plus revision log.

CALIBRATION NOTE

Score honestly. A "3" means the student met the standard the way an adult professional would. A "4" means they did something a working entrepreneur would be proud to show. Withhold "4" unless the work earns it.

SENSITIVITY AND FAMILY ENGAGEMENT

What We Don't Ask. What We Always Frame.

Money is a tool, not a virtue. Some students live in households with credit-card debt, low or no banking access, or active financial stress. The curriculum is built so that none of those students is asked to disclose, perform, or apologize for their family situation.

Rules of the room

- **Never ask about family money.** No questions about parents income, debts, accounts, or banking status.
- **Use neutral hypotheticals.** "For a sample family with a $4,000/month take-home" — not "your family."
- **Decouple wealth from worth.** Cash never says "everyone can be rich." Cash says "you will be a clearer thinker about money no matter what you end up doing."
- **Name systemic factors honestly.** Wage gaps, geography, and luck shape outcomes. The skill we build is the skill in the student hands.
- **Protect the venture.** A student Capstone is theirs. We don't require they share revenue or family contribution. The artifact is the work, not the bank account.

Family engagement

Parents get one well-framed question per quarter — never a quiz, never a money-disclosure prompt. The default frame is curiosity ("What did you notice about your customers this week?"), not interrogation ("How much did you make?").

"Money is a tool. The question is not whether your family has a lot of it. The question is what you do with the moves we are practicing — interview, math, message, listen — wherever your starting line happens to be."

— CASH

ACTIVITY INDEX

Ten Activities. One Notebook.

Every activity in this book is built to be done with a pencil, your own evidence, and the questions Cash keeps asking. Skip nothing. Date everything. Re-read in three months.

No.	Activity	What you will do
01	Day 1 Customer Assumption	One sentence, one card. Don't try to be right; try to be specific.
02	Open vs. Leading Questions	Rewrite five leading questions into open ones. Practice on a partner.
03	Coding Patterns from Transcripts	Highlight surprises, frustrations, wishes. Tally them. Find the pattern.
04	Persona Pack v1	Combine evidence into one specific person — name, behaviors, top frustration.
05	Variable Cost per Unit	From a real receipt, compute the cost per one of the thing you sell.
06	Contribution Margin Worksheet	Price minus variable cost. Decide if it covers your fixed costs.
07	Two-Message A/B Test	Two honest messages. Three-day test. Report the winner and why.
08	Persuasion vs. Manipulation Line	Sort 10 ad claims onto a line. Defend your placement.
09	1-Week Feedback Loop	Collect, code, and act on feedback. Document three changes.
10	Final Capstone Defense	Five slides. Five minutes. Show the moves you made.

NOTEBOOK SETUP

Get a single notebook for the whole year. Number every page. The first 5 pages are your Vocabulary Spine. The last 5 pages are your Day-1 Assumptions (you will come back to those). Everything in between is your work.

Notebook checklist

- **Title page** — your name, your venture working-title, today's date.
- **Vocabulary spine** — 5 pages reserved for definitions you will add as we go.
-

Customer notes — one section for interview transcripts. Use direct quotes, with initials and dates.

- **Math section** — receipts taped in, per-unit computations, contribution margin scratch work.
- **Messaging section** — drafts of two-claim A/B tests, with what won and why.
- **Feedback log** — one page per week of your feedback system, with the three changes you made.

WORKED EXAMPLE

Worked Example: Per-Customer Math at the Snack Stand

Devon sells protein cookies after school. He sells 30 cookies a week at $2.00 each. Revenue is $60. He spends $0.85 on each cookie (oats, peanut butter, packaging). His variable cost per cookie is $0.85.

Contribution margin per cookie = $2.00 - $0.85 = **$1.15**. His fixed cost (the table he rents from the school store) is $20/week. He needs **$20 / $1.15 = 18 cookies/week** just to cover the table. He's currently selling 30. He's clearing $14.50/week after fixed costs.

Devon's question to Cash: should he raise the price to $2.50? Cash answers with a question: "Which people, and how do you know? Run a 3-day test. Half the kids see $2.00. Half see $2.50. Tell me what they actually buy."

WHAT TO COPY

The move here is not the answer. It is the structure: specific number, specific person, specific decision. Use that pattern in your own venture this quarter.

> *"Most adults skip this. You won't. Pick one customer. Compute one number. Decide one thing." — Cash*

END-OF-QUARTER REFLECTION

Four Quarters. Same Five Questions.

At the end of every quarter you will answer the same five questions. The answers should look different over time. Reread the previous quarter before you write a new one.

1. **Which people did you talk to this quarter, and what did they actually say?**

   Specific quotes, initials, dates. Resist the urge to summarize.
2. **What did you change in your venture because of evidence?**

   Name the change. Name the evidence that caused it.
3. **What is one number you know better now than three months ago?**

   Variable cost per unit. Contribution margin. Anything quantitative.
4. **What is a claim you made that you would take back?**

   Be honest. The claim does not have to be public. The honesty is the move.
5. **What would Cash ask you about all of this?**

   "Which people, and how do you know?" is always available. Find your own version.

> *"Reflection is not a feeling check-in. It is a memory tool. The you in three months wants to know what the you today learned. Write it down so future-you can use it."*
>
> — CASH

One promise.

Cash to Company makes one promise: students who move through this program will be clearer thinkers about money and entrepreneurship — no matter what they end up doing for a living.

We do not promise wealth. We do not promise success. We promise clarity.

"Customers are evidence, not assumptions."

— CASH, GRADE 7